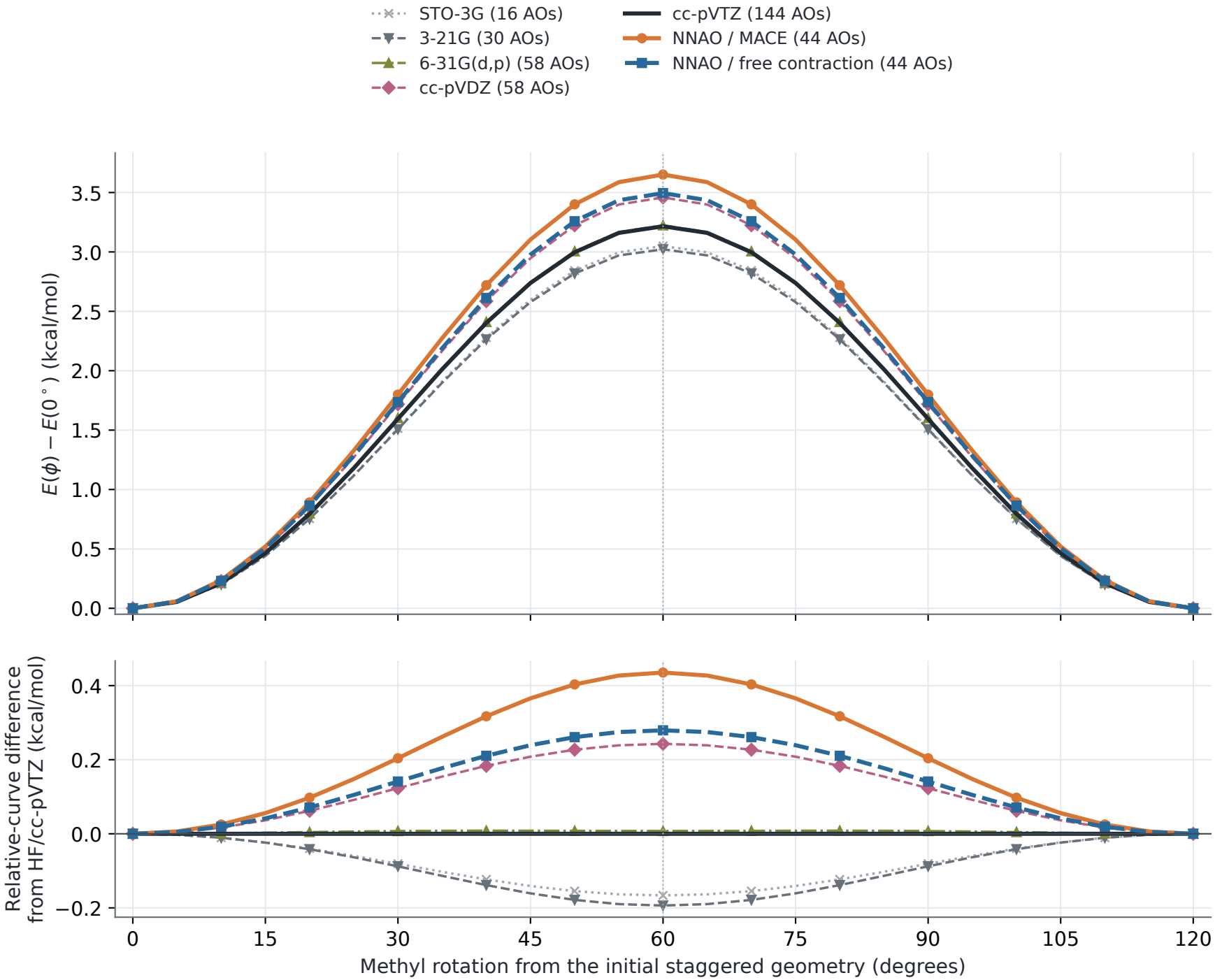
Ethane torsion: NNAO and conventional SZ-to-TZ basis sets
STO-3G (16 AOs)
3-21G (30 AOs)
6-31G(d,p) (58 AOs)
cc-pVDZ (58 AOs)
cc-pVTZ (144 AOs)
NNAO / MACE (44 AOs)
NNAO / free contraction (44 AOs)
$E(\phi) - E(0^\circ)$ (kcal/mol)
Relative-curve difference from HF/cc-pVTZ (kcal/mol)
Methyl rotation from the initial staggered geometry (degrees)
25 rigid geometries, 5-degree spacing; DF-RHF / def2-universal-jkfit, spherical AOs.
Each curve uses its own 0-degree energy. HF/cc-pVTZ is a comparison basis, not a complete-basis reference.